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CANDIDATE NUMBER

SYDNEY GRAMMAR SCHOOL



2016 Trial Examination

FORM VI

MATHEMATICS 2 UNIT

Tuesday 9th August 2016

General Instructions

- Reading time — 5 minutes
- Writing time — 3 hours
- Write using black pen.
- Board-approved calculators and templates may be used.

Total — 100 Marks

- All questions may be attempted.

Section I – 10 Marks

- Questions 1–10 are of equal value.
- Record your answers to the multiple choice on the sheet provided.

Section II – 90 Marks

- Questions 11–16 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each answer booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Write your candidate number on this question paper and hand it in with your answers.
- Place everything inside the answer booklet for Question Eleven.

Checklist

- SGS booklets — 6 per boy
- Multiple choice answer sheet
- Reference sheet
- Candidature — 88 boys

Examiner

PKH

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

What are the solutions of $x^2 - 3x + 1 = 0$?

- (A) $x = \frac{3 \pm \sqrt{5}}{2}$
- (B) $x = \frac{-3 \pm \sqrt{13}}{2}$
- (C) $x = \frac{3 \pm \sqrt{13}}{2}$
- (D) $x = \frac{-3 \pm \sqrt{5}}{2}$

QUESTION TWO

What is the limiting sum for the infinite geometric series $12 - 6 + 3 - \dots$?

- (A) 24
- (B) 8
- (C) -8
- (D) -12

QUESTION THREE

What is the derivative of $\frac{2}{x}$?

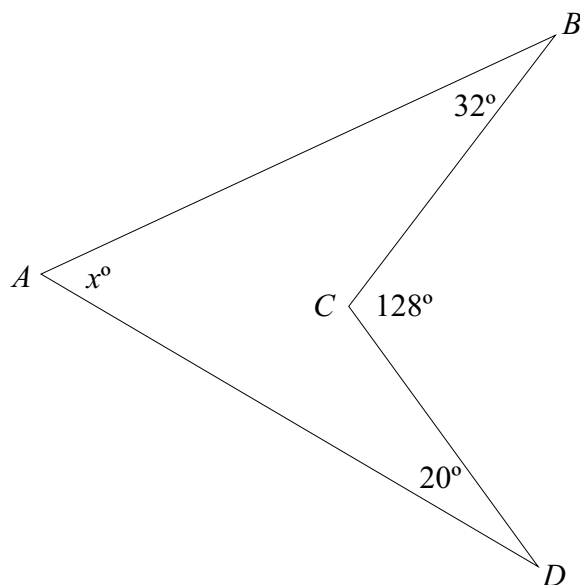
- (A) $2 \ln x$
- (B) $\ln 2x$
- (C) $-\frac{2}{x^2}$
- (D) $\frac{2}{x^2}$

QUESTION FOUR

Which of the following is a primitive of e^{2x} ?

- (A) $(2x + 1)e^{2x+1}$
- (B) $2e^{2x}$
- (C) $\frac{e^{2x+1}}{2x + 1}$
- (D) $\frac{e^{2x}}{2}$

QUESTION FIVE



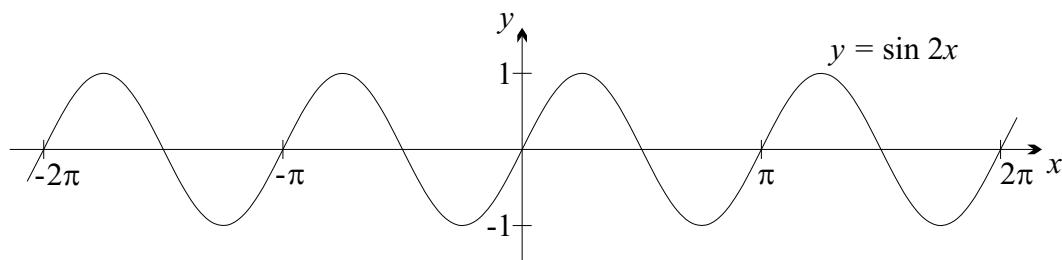
What is the value of x in the diagram above?

- (A) 66
- (B) 76
- (C) 64
- (D) 86

QUESTION SIX

Simplify $\log_4 54 - 2\log_4 3$.

- (A) $\log_4 9$
- (B) $\log_4 48$
- (C) $\log_4 6$
- (D) 1

QUESTION SEVEN

The graph of $y = \sin 2x$ is drawn. How many solutions does the equation $\frac{1}{6}x = \sin 2x$ have?

- (A) 3
- (B) 4
- (C) 7
- (D) 8

QUESTION EIGHT

Consider the points $A(1, -2)$ and $B(3, 6)$. What is the equation of the perpendicular bisector of AB ?

- (A) $y - 2 = -\frac{1}{4}(x - 2)$
- (B) $y - 2 = 4(x - 2)$
- (C) $y - 4 = -1(x - 1)$
- (D) $y + 2 = -\frac{1}{4}(x - 1)$

QUESTION NINE

What is the greatest value of $\frac{20}{4\sin^2\theta + 2\cos^2\theta}$ for $0 \leq \theta \leq \frac{\pi}{2}$?

- (A) 10
- (B) 5
- (C) 20
- (D) $\frac{20}{6}$

QUESTION TEN

Which of the following is a correct simplification of $\frac{\cos(\pi - x)}{\cos\left(\frac{\pi}{2} - x\right)}$?

- (A) $\cos \frac{\pi}{2}x$
- (B) $-\tan x$
- (C) $-\cot x$
- (D) $\tan x$

————— End of Section I —————

Examination continues overleaf ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION ELEVEN (15 marks) Use a separate writing booklet. **Marks**

- | | |
|---|---|
| (a) Calculate $3e^{1.5}$ correct to 3 decimal places. | 1 |
| (b) Find the gradient of the line $3y - 2x = 6$. | 1 |
| (c) Factorise $9a^2 - 16$. | 1 |
| (d) Differentiate $x^3 e^x$. | 2 |
| (e) Differentiate $(3 + \sin x)^4$. | 2 |
| (f) Solve the inequation $5 - 2x \geq 14$. | 2 |
| (g) Solve $ 2x - 5 = 7$. | 2 |
| (h) Find the coordinates of the focus of the parabola $(x - 2)^2 = 8y + 16$. | 2 |
| (i) Solve $2 \sin \theta = -1$ for $0 \leq \theta \leq 2\pi$. | 2 |

QUESTION TWELVE (15 marks) Use a separate writing booklet. **Marks**

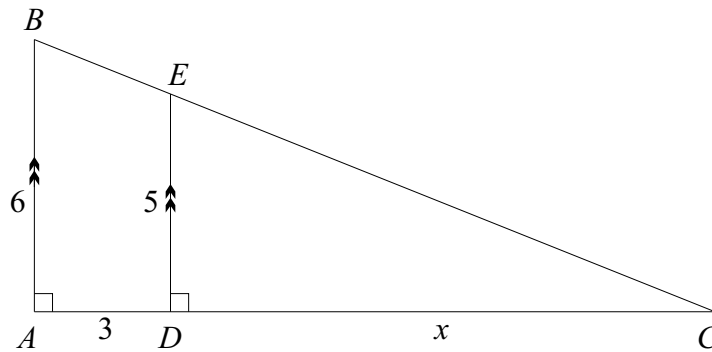
- | | |
|---|---|
| (a) Make y the subject of the equation $x = \log_3 y$. | 1 |
| (b) Find $\int \frac{4x^3}{2 + x^4} dx$. | 1 |
| (c) Differentiate $\frac{x}{\sin x}$. | 2 |
| (d) Evaluate $11 + 16 + 21 + \dots + 101$. | 3 |
| (e) The quadrilateral $ABCD$ has vertices $A(0, 4)$, $B(4, 8)$, $C(-1, -4)$ and $D(-5, -8)$. | |
| (i) Show that $ABCD$ is a parallelogram. | 2 |
| (ii) Find the equation of line BC , leaving your answer in the form $ax + by + c = 0$. | 2 |
| (iii) Find the perpendicular distance from A to line BC . | 2 |
| (iv) Find distance BC . | 1 |
| (v) Hence find the area of $ABCD$. | 1 |

Examination continues next page ...

QUESTION THIRTEEN (15 marks) Use a separate writing booklet.

Marks

(a)



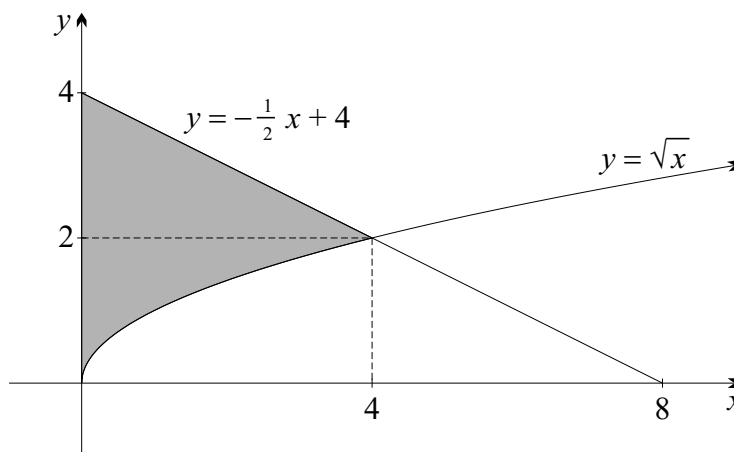
(i) Prove that $\triangle ABC \parallel \triangle DEC$ in the diagram above.

1

(ii) Find the value of x , giving reasons.

2

(b)



Find the shaded area in the diagram above.

3

(c) A person walks on the true bearing of 050° for 20km from point P and stops at point A . Another person walks for 30km on a bearing of 110° from point P and stops at point B .

(i) Represent this information on a neat diagram.

1

(ii) Find the distance AB to the nearest kilometre.

2

(iii) Find the bearing of A from B to the nearest degree.

2

(d) The volume V is the number of litres of water in a tank at time t minutes. Water is flowing into the tank at a rate given by $\frac{dV}{dt} = \frac{4}{2t+1}$ litres per minute. At time $t = 0$ the water begins to flow into an empty tank. How much water is in the tank after 5 minutes, to the nearest tenth of a litre?

2

(e) Use the trapezoidal rule with 3 function values to estimate $\int_1^3 2^x dx$.

2

Examination continues overleaf ...

QUESTION FOURTEEN (15 marks) Use a separate writing booklet.

Marks

(a) Differentiate $\log_e(e^x + 2)$.

2

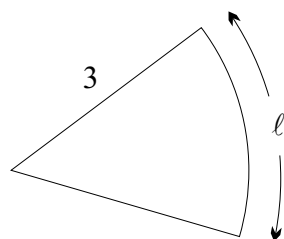
(b) A sum of \$20 000 is invested at a fixed rate of interest, compounded annually. After 5 years the principal has grown to \$28 567.

3

Find the annual rate of interest to the nearest tenth of one percent.

(c)

2



The sector, shown in the diagram above, has an area of 36 square units and a radius of 3 units. Find the arc length ℓ .

(d) Solve the equation $\tan^2 \theta + \sqrt{3} \tan \theta = 0$ for $0 \leq \theta \leq 2\pi$.

2

(e) A particle is moving in a straight line with velocity given by $\dot{x} = 3t^2 - 9t$ where t is measured in seconds and x is measured in metres. Its displacement from the origin is initially 10 metres.

(i) Find the displacement x as a function of t .

2

(ii) Find the displacement when the acceleration is zero.

2

(iii) Find the average speed during the first 4 seconds.

2

QUESTION FIFTEEN (15 marks) Use a separate writing booklet.**Marks**

- (a) Find the volume formed when $y = \sec 2x$ is rotated about the x -axis from $x = 0$ to $x = \frac{\pi}{8}$. **2**
- (b) Find $\int (\sqrt[3]{x-9})^2 dx$. **2**
- (c) The population P of a town is growing at a rate proportional to its size at any time, so that $\frac{dP}{dt} = kP$, for some constant k . At the beginning of 2010 the town's population was 23 000 and at the beginning of 2016 its population had grown to 28 000.
- (i) Show that $P = Ae^{kt}$ satisfies the equation $\frac{dP}{dt} = kP$. **1**
- (ii) Find the value of A . **1**
- (iii) Find the value of k . **2**
- (iv) Estimate, to the nearest hundred, what the population will be at the beginning of 2025. **1**
- (v) During which year will the population be double the size it was at the beginning of 2010? **2**
- (d) A person borrows \$400 000 and makes regular monthly repayments of \$ M . The interest rate is 6% per annum compounded monthly. The loan is taken over a period of 20 years. Let A_n be the amount owing after n months, just after a repayment has been made.
- (i) Find an expression for A_2 . **1**
- (ii) Find the monthly payment M to the nearest cent. **3**

QUESTION SIXTEEN (15 marks) Use a separate writing booklet.**Marks**(a) Consider the function $y = x^5 - 80x$.(i) Find the x -intercepts.**1**

(ii) Find the stationary points and determine their nature.

2

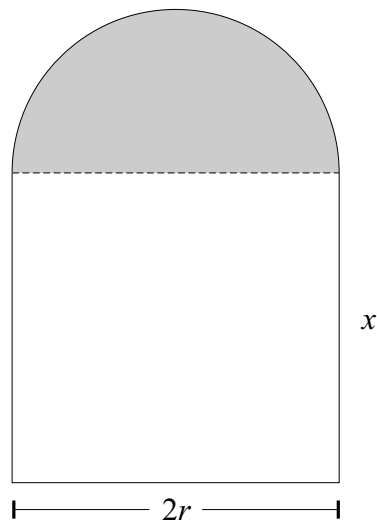
(iii) Find the point of inflexion.

2

(iv) Draw a neat sketch of the function, showing the above information.

2

(b)



A large window is constructed in the shape of a rectangle with a semicircle on top, as in the diagram above. The glass forming the semicircle is opaque and the glass forming the rectangle is clear. The height of the rectangle is x metres and the radius of the semicircle is r metres. The perimeter of the entire window is 12 metres.

(i) Show that $x = 6 - \frac{\pi}{2}r - r$.**2**

(ii) The window is constructed so that the area of the rectangle, made of clear glass, is maximised.

3Show that $r = \frac{6}{\pi + 2}$.(c) The cubic function $y = ax^3 + bx^2 + cx + d$ has two stationary points and one point of inflexion.**3**

Prove that the x -coordinate of the point of inflexion is located at the average of the x -coordinates of the two stationary points.

————— End of Section II —————

END OF EXAMINATION

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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Question Nine

A ☐ B ☐ C ☐ D ☐

Question Ten

A ☐ B ☐ C ☐ D ☐

Question 1

$$x^2 - 3x + 1 = 0$$

$$x = \frac{3 \pm \sqrt{9-4}}{2}$$

$$x = \frac{3 \pm \sqrt{5}}{2}$$

(A)

$$6 \log_4 54 - 2 \log_4 3$$

$$= \log_4 54^6 - \log_4 3^2$$

$$= \log_4 6 \quad (C)$$

Question 2

$$S_{\infty} = \frac{a}{1-r}$$

$$S_{\infty} = \frac{12}{1-\frac{1}{2}}$$

$$= 12 \times \frac{2}{1}$$

$$= 8 \quad (B)$$

Question 3

$$y = 2x^{-1}$$

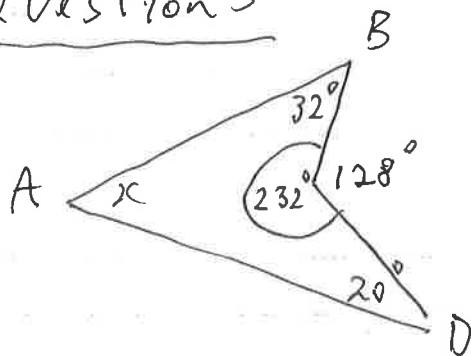
$$y' = -2x^{-2}$$

$$= -\frac{2}{x^2} \quad (C)$$

Question 4

$$\int e^{2x} dx = \frac{1}{2} e^{2x} + C \quad (D)$$

Question 5



$$32 + 20 + 232 + x = 360^\circ$$

$$284 + x = 360$$

$$x = 76 \quad (B)$$

$$1. \text{ The line } y = \frac{1}{6}x$$

when drawn carefully cuts the curve 7 times. (C)

$$8 \quad m = \frac{6-2}{3-1} = 4$$

$$m_{\perp} = -\frac{1}{4} \quad M = (2, 2)$$

Equ is

$$y + 2 = -\frac{1}{4}(x - 1)$$

(A)

$$9. \quad 4 \sin^2 \theta + 2 \cos^2 \theta$$

$$= 2 \sin^2 \theta + 2 \cos^2 \theta + 2 \sin^2 \theta$$

$$= 2 + 2 \sin^2 \theta$$

Least when $\sin \theta = 0$

$$\text{Max of expression is } \frac{20}{2} = 10 \quad (A)$$

$$10 \quad \frac{\cos(\pi - x)}{\cos(\frac{\pi}{2} - x)}$$

$$= \frac{-\cos x}{\sin x}$$

$$= -\cot x$$

(C)

QUESTION ELEVEN

(2)

$$(a) \quad 3e^{1.5} \quad \checkmark \\ = 13.445 \quad (3 \text{ de})$$

$$(b) \quad 3y - 2x = 6 \\ y = \frac{2}{3}x + 2$$

Gradient is $\frac{2}{3}$ $\checkmark$

$$(c) \quad 9a^2 - 16 \\ = (3a - 4)(3a + 4) \quad \checkmark$$

$$(d) \quad y = \underset{u}{x^3} \underset{v}{e^x}$$

$$y' = uv' + v u' \\ = x^3 e^x + 3x^2 e^x \quad \checkmark \checkmark \\ = x^2 e^x (x + 3)$$

$$(e) \quad y = (3 + \sin x)^4 \\ y' = 4(3 + \sin x)^3 \cos x \quad \checkmark$$

$$(f) \quad 5 - 2x \geq 14 \\ -2x \geq 9 \quad \checkmark \\ x \leq -4.5 \quad \checkmark$$

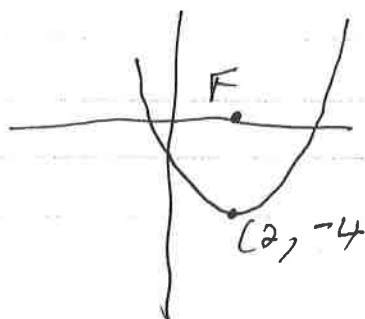
$$(g) \quad |2x - 5| = 7$$

$$2x - 5 = 7 \quad \text{OR} \quad 2x - 5 = -7$$

$$x = 6 \quad \text{OR} \quad x = -1 \\ \checkmark \quad \checkmark$$

$$(h) \quad (x-2)^2 = 4(y+4)$$

$$a = 4, \quad V = (2, -4) \quad \checkmark$$



$$F = (2, 0) \quad \checkmark$$

$$(i) \quad 2 \sin \theta = -1$$

$$\sin \theta = -\frac{1}{2}$$

$$\theta = \pi + \frac{\pi}{6} \quad \text{OR} \quad 2\pi - \frac{\pi}{6} \\ = \frac{7\pi}{6} \quad \text{OR} \quad \frac{11\pi}{6} \\ \checkmark \quad \checkmark$$

Question 12

(3)

(a) $x = \log_3 y$

$$y = 3^x \quad \checkmark$$

(b) $\int \frac{4x^3}{2+x^4} dx$
 $= \ln(2+x^4) + C \quad \checkmark$

(c) $y = \frac{x}{\sin x} - u$
 $\sin x = v$

$$y' = \frac{vu' - uv'}{v^2}$$

$$= \frac{\sin x \cdot 1 - x \cos x}{\sin^2 x} \quad \checkmark$$

(d)

$$T_n = 101$$

$$a + (n-1)d = 101$$

$$11 + (n-1)5 = 101 \quad \checkmark$$

$$11 + 5n - 5 = 101$$

$$5n = 95$$

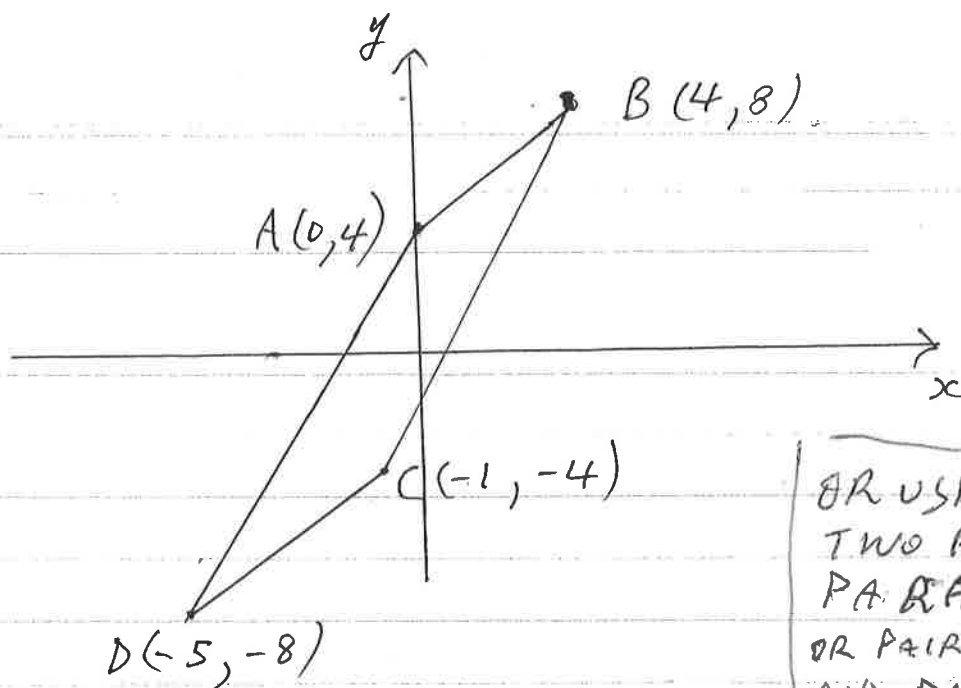
$$n = 19 \quad \checkmark$$

$$S_n = \frac{n}{2}(a+l)$$

$$S_{19} = \frac{19}{2}(11+101)$$

$$= 1064 \quad \checkmark$$

(e)



(4)

OR USE
TWO PAIRS OF
PARALLEL SIDES
OR PAIR OF EQUAL
AND PARALLEL SIDES

(i) Let M = mid point of AC .

$$M = \left(\frac{0 + (-1)}{2}, \frac{4 + (-4)}{2} \right) = \left(-\frac{1}{2}, 0 \right)$$

$$\text{Mid point of } BD = \left(-\frac{1}{2}, 0 \right)$$

So diagonals bisect each other ✓

So $ABCD$ is a parallelogram

(ii) $m(BC) = \frac{8 - (-4)}{4 - (-1)} = \frac{12}{5}$

Eqn of line BC is

$$y + 4 = \frac{12}{5}(x + 1) \quad \checkmark$$

$$5y + 20 = 12x + 12 \quad \checkmark$$

$$12x - 5y - 8 = 0 \quad \checkmark$$

(iii) $d_{\perp} = \left| \frac{Ax_1 + By_1 + C}{\sqrt{A^2 + B^2}} \right|$ ✓

$$= \frac{|12 \times 0 - 5 \times 4 - 8|}{\sqrt{12^2 + (-5)^2}} = \frac{28}{13} \quad \checkmark$$

(iv) $d(BC) = \sqrt{(4 - (-1))^2 + (8 - (-4))^2}$

$$= 13 \quad \checkmark$$

(v) $\text{Area} = b \times h$

$$= 13 \times \frac{28}{13} = 28 \quad \checkmark$$

Question 13

5

a (i) $\angle ACB$ is common

$$\angle BAC = \angle EDC = 90^\circ$$

$\triangle ABC \parallel \triangle DEC$ (AAA)

Must have this.

(ii)

$$\frac{x}{x+3} = \frac{5}{6} \quad \left(\begin{array}{l} \text{corresponding} \\ \text{matching sides in} \\ \text{similar } \triangle's \end{array} \right)$$

$$6x = 5x + 15$$

$$x = 15$$

(b)

$$A = \int_a^b y_1 - y_2 \, dx$$

$$= \int_0^4 -\frac{1}{2}x + 4 - x^{\frac{1}{2}} \, dx$$

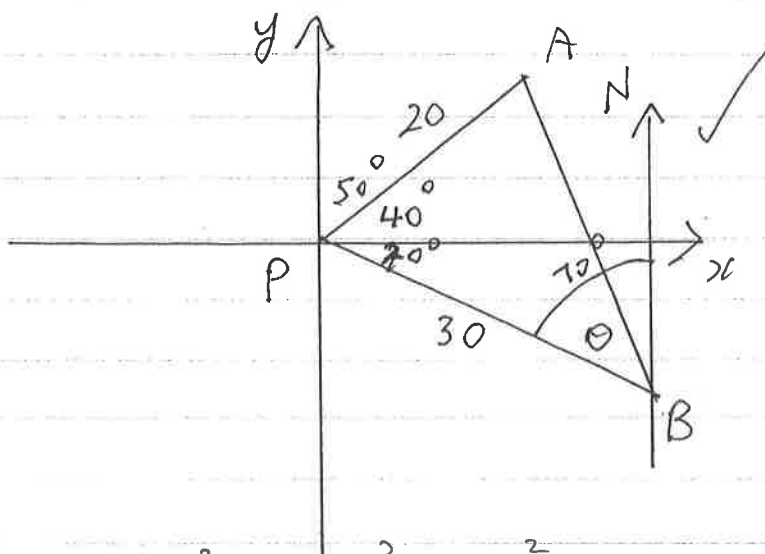
$$= \left[-\frac{x^2}{4} + 4x - \frac{2}{3}x^{\frac{3}{2}} \right]_0^4$$

$$= -\frac{16}{4} + 16 - \frac{2}{3} \times 8 - [0]$$

$$= \frac{20}{3} \, \text{a}^2$$

(c)

(i)



$$\angle APB = 60^\circ$$

$$\cos 60^\circ = \frac{1}{2}$$

(ii)

$$AB^2 = 20^2 + 30^2 - 2 \times 20 \times 30 \cos 60^\circ$$

$$AB^2 = 1300 - 600$$

$$AB = \sqrt{700} \doteq 26.4575$$

$$= 26 \, \text{km (to nearest km)}$$

(iii)

$$\cos \theta = \frac{30^2 + (26.4575)^2 - 20^2}{2 \times 30 \times 26.4575} \quad \checkmark \quad 6$$

$$\theta = 40.89 = 41^\circ \text{ to nearest degree}$$

$$\begin{aligned} \text{Bearing} &= 360^\circ - 70 + \theta \\ &= 331^\circ \text{ (nearest degree)} \quad \checkmark \end{aligned}$$

(iv)

$$\frac{dV}{dt} = \frac{4}{2t+1}$$

$$V = 4 \ln(2t+1) + C$$

$$V = 2 \ln(2t+1)^2 + C \quad \checkmark$$

When

$$t=0$$

$$V=0$$

$$\left. \begin{array}{l} t=0 \\ V=0 \end{array} \right\} \begin{array}{l} 0 = 2 \ln 1 + C \\ C = 0 \end{array}$$

$$V = 2 \ln(2t+1)$$

Must show calculation of C

$$\text{When } t=5 \quad V = 2 \ln 11$$

$$= 4.795 \dots \quad \checkmark$$

$$= 4.8 \text{ L (to nearest tenth of a litre)}$$

(v)

x	1	2	3
y	2	4	8

✓

$$\begin{aligned} \int_1^3 2^x dx &= \frac{2-1}{2} (2+4) + \frac{3-2}{2} (4+8) \\ &= 9 \quad \checkmark \end{aligned}$$

Question 14

7

(a) $y = \ln(e^{2x} + 2)$

$$y' = \frac{e^{2x}}{e^{2x} + 2} \quad \checkmark$$

(b) $P = A \left(1 + \frac{r}{100}\right)^n$

$$28567 = 2000 \left(1 + \frac{r}{100}\right)^5 \quad \checkmark$$

$$\left(1 + \frac{r}{100}\right)^5 = \frac{28567}{20000}$$

$$1 + \frac{r}{100} = \sqrt[5]{\frac{28567}{20000}} \quad \checkmark$$

$$1 + \frac{r}{100} = 1.0739 \dots$$

$$\frac{r}{100} = 0.0739 \dots$$

$$r = 7.39 \dots$$

So rate is 7.4% $\checkmark$

(c) $A = \frac{1}{2} r^2 \theta = 36$

$$\frac{1}{2} \times 9 \times \theta = 36$$

$$\theta = 8 \quad \checkmark$$

$$l = r\theta$$

$$= 3 \times 8$$

$$= 24 \text{ units}$$

(d) $\tan^2 \theta + \sqrt{3} \tan \theta = 0$ for $0 \leq \theta < 2\pi$

$$\tan \theta (\tan \theta + \sqrt{3}) = 0$$

$$\tan \theta = 0 \text{ or } \tan \theta = -\sqrt{3}$$

$$\theta = 0, \pi, 2\pi \text{ or } \theta = \frac{2\pi}{3}, \frac{5\pi}{3}$$

$\checkmark$

$\checkmark$

(8)

(e) (i) $\dot{x} = 3t^2 - 9t$

$$x = t^3 - \frac{9}{2}t^2 + C \quad \checkmark$$

$t=0 \quad 10 = 0 - 0 + C$

$x=10$

$C = 10 \quad \checkmark$

$$x = t^3 - \frac{9}{2}t^2 + 10$$

(ii)

$$\ddot{x} = 6t - 9$$

$\ddot{x} = 0 \quad t = \frac{3}{2}$

When $t = \frac{3}{2}$ $x = \left(\frac{3}{2}\right)^3 - \frac{9}{2} \times \left(\frac{3}{2}\right)^2 + 10$

$$x = \frac{27}{8} - \frac{81}{8} + \frac{80}{8}$$

$$x = \frac{26}{8} = \frac{13}{4}$$

(iii) The particle can change direction when $\dot{x} = 0$

$$3t^2 - 9t = 0$$

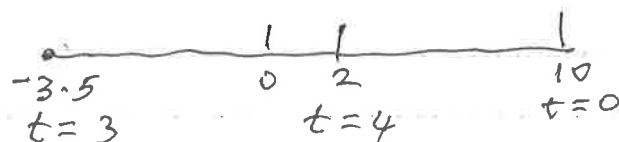
$$3t(t-3) = 0$$

$$t = 0 \text{ or } t = 3 \quad \checkmark$$

$t=0, \quad x=10$

$t=3, \quad x = 27 - \frac{9}{2} \times 9 + 10 = -3.5$

$t=4, \quad x = 64 - \frac{9}{2} \times 16 + 10 = 2$



Total distance travelled = $13.5 + 5.5 = 19$

Average speed = $\frac{19}{4} = 4.75 \text{ m/s} \quad \checkmark$
over first 4 secs

Question 15

9

$$\begin{aligned} (a) \quad V &= \pi \int_0^{\frac{\pi}{8}} \sec^2 2x \, dx \\ &= \pi \left[\frac{\tan 2x}{2} \right]_0^{\frac{\pi}{8}} \checkmark \\ &= \frac{\pi}{2} \left[\tan \frac{\pi}{4} - \tan 0 \right] \\ &= \frac{\pi}{2} \text{ unit}^2 \checkmark \end{aligned}$$

$$\begin{aligned} (b) \quad &\int (3\sqrt{x-9})^2 \, dx \\ &= \int (x-9)^{\frac{2}{3}} \, dx \checkmark \\ &= \frac{(x-9)^{\frac{5}{3}}}{\frac{5}{3}} + C \\ &= \frac{3}{5} (x-9)^{\frac{5}{3}} + C \checkmark \end{aligned}$$

$$\begin{aligned} (c) \quad (i) \quad P &= A e^{kt} \\ \frac{dP}{dt} &= k A e^{kt} \\ \frac{dP}{dt} &= k P \checkmark \end{aligned}$$

$$\begin{aligned} (ii) \quad P &= A e^{kt} \\ \text{In 2010, } t=0 \quad 23000 &= A e^0 \checkmark \\ P &= 23000 \quad A = 23000 \end{aligned}$$

$$\begin{aligned} (iii) \quad \text{In 2016 } t=6 \quad 28000 &= 23000 e^{6k} \checkmark \\ P &= 28000 \\ K &= \frac{1}{6} \ln\left(\frac{28}{23}\right) \checkmark \\ K &= 0.032785... \end{aligned}$$

(iv) $P = 23000 e^{15K}$

$P \div 37600$ (to nearest hundred) ✓

(v) $P = 23000 e$

$t = ?$ $46000 = 23000 e^{Kt}$

$P = 46000$
(doubled)

$e^{Kt} = 2$

$Kt = \ln 2$

$t = \frac{1}{K} \ln 2$

$t = 21.14 \dots$ ✓

Doubles during the 22nd year
i.e. 2031.

(c) (i) $A_1 = 400,000 \times 1.005 - M$ 6% pa
 $= 0.005$

$A_2 = (400,000 \times 1.005 - M) 1.005 - M$ monthly

* Must show 3 terms.

$A_2 = 400,000 \times 1.005^2 - M(1 + 1.005)$ *

(ii) $A_n = 400,000 \times 1.005^n - M(1 + 1.005 + 1.005^2 + \dots + 1.005^{n-1})$

$A_{240} = 400,000 \times (1.005)^{240} - M(1 + 1.005 + \dots + 1.005^{23})$

But $A_{240} = 0$ ✓

So $400,000 \times 1.005^{240} = M \frac{(1.005^{240} - 1)}{1.005 - 1}$ ✓

$400,000 \times 1.005^{240} = M \times \frac{(1.005^{240} - 1)}{0.005}$

$M = \$2865.72$ ✓

Question 16

11

(a) $y = x^5 - 80x$

(i) x intercepts where $y = 0$

$$x^5 - 80x = 0$$

$$x(x^4 - 80) = 0$$

$$x = 0, \sqrt[4]{80}, -\sqrt[4]{80} \quad \checkmark$$

(ii) $y' = 5x^4 - 80$

stat pts where $y' = 0$

$$5x^4 = 80$$

$$x^4 = 16$$

$$x = \pm 2 \quad \checkmark$$

$$y'' = 20x^3$$

When $x = 2, y'' = 160 > 0$

Min pt at $(2, -128)$

When $x = -2, y'' = -160 < 0$

Max pt at $(-2, 128)$ } $\checkmark$

(iii) Possible point of inflexion where $y'' = 0$

Table of values for y''

at $x = 0$

x	-1	0	1
y''	-20	0	20

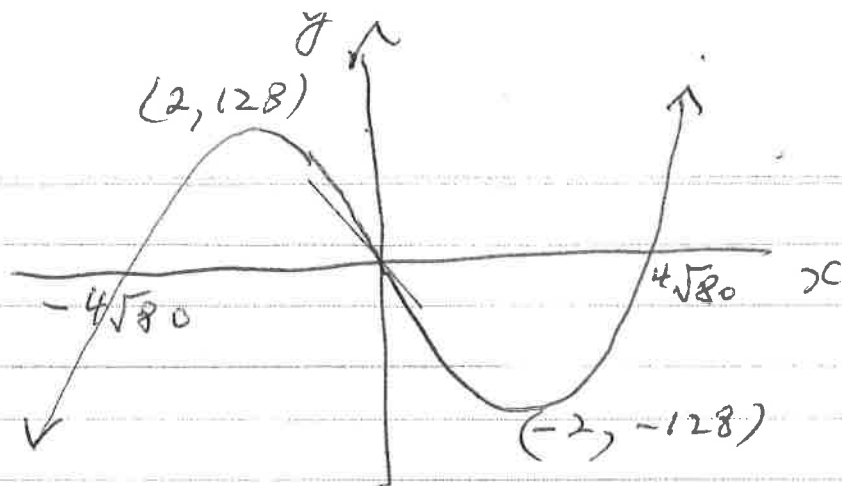
 $\checkmark$

There is a change in concavity

at $x = 0 \quad \checkmark$

So $(0, 0)$ is a point of inflexion

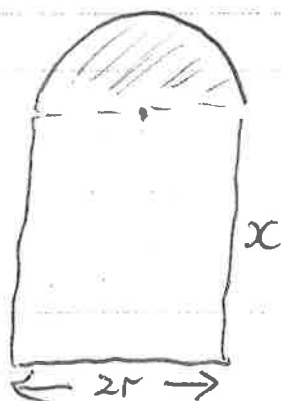
(IV)



(12)

(b)

(i)



$$P = 2r + \pi r + 2x \quad \checkmark$$

$$12 = 2r + \pi r + 2x \quad \checkmark$$

$$x = 6 - \frac{\pi}{2}r - r$$

(ii)

$$A = 2r x$$

$$= 2r \left(6 - \frac{\pi}{2}r - r \right)$$

$$A = 12r - \pi r^2 - 2r^2$$

$$A' = 12 - 2\pi r - 4r \quad \checkmark$$

$$A'' = -2\pi - 4 < 0 \quad \checkmark$$

Max value where $A' = 0$

$$12 - 2\pi r - 4r = 0$$

$$6 - \pi r - 2r = 0$$

$$6 = \pi r + 2r$$

$$r = \frac{6}{\pi + 2}$$

(c) $y = ax^3 + bx^2 + cx + d$

$$y' = 3ax^2 + 2bx + c$$

Let α co-ords of the stationary pts be α and β

α and β are roots of

$$3ax^2 + 2bx + c = 0$$

$$\alpha + \beta = \Sigma \text{ roots}$$

$$\alpha + \beta = -\frac{2b}{3a}$$

$$\text{Average of } \alpha \text{ and } \beta = \frac{\alpha + \beta}{2} = -\frac{b}{3a}$$

We are told that there is a point of inflexion.

This occurs when $y'' = 0$

$$6ax + 2b = 0$$

$$x = -\frac{b}{3a}$$

which is the average of α and β .